

Q: 2(i)

A: 2(i)

(a) atomic number: 15  
mass number: 31

number of protons: 15  
number of electrons: 15  
number of neutrons: 16

(b) Subshells that are present in N-shell:

4s, 4p, 4d, 4f

f-sub-shell can hold a maximum of 14 electrons.

Q: 3(ii)

A: 3(ii)

(a) four pairs

| Pair     | Electronic C                                                     | Groups                         |
|----------|------------------------------------------------------------------|--------------------------------|
| A: Cs, H | A: $1s^2, 2s^2$ (Be), H: $1s^2, 2s^2, 2p^6, 3s^2$ (Mg)           | Group 2 (alkaline earth metal) |
| B: C, D  | B: $1s^2, 2s^2, 2p^6$ (Ne)<br>D: $1s^2$ (He)                     | group 18 (noble gas)           |
| C: C, E  | C: $1s^2, 2s^2, 2p^3$ (N), E: $1s^2, 2s^2, 2p^4, 3s^2, 3p^1$ (P) | Group 15 (VA)                  |

|        |                                                            |                          |
|--------|------------------------------------------------------------|--------------------------|
| F Cg G | $F = 1s^2, 2s^1 (Li)$<br>$G = 1s^2, 2s^2, 2p^6, 3s^1 (Na)$ | group :- 1 alkali metals |
|--------|------------------------------------------------------------|--------------------------|

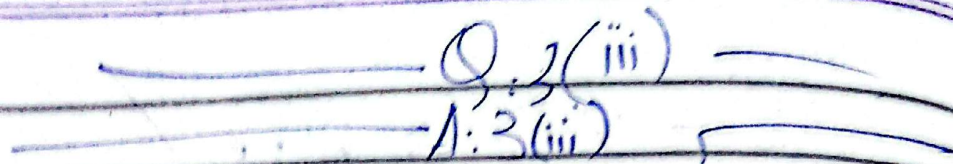
⑥ Classification :-

A Cg H (Be, Mg) = metals.

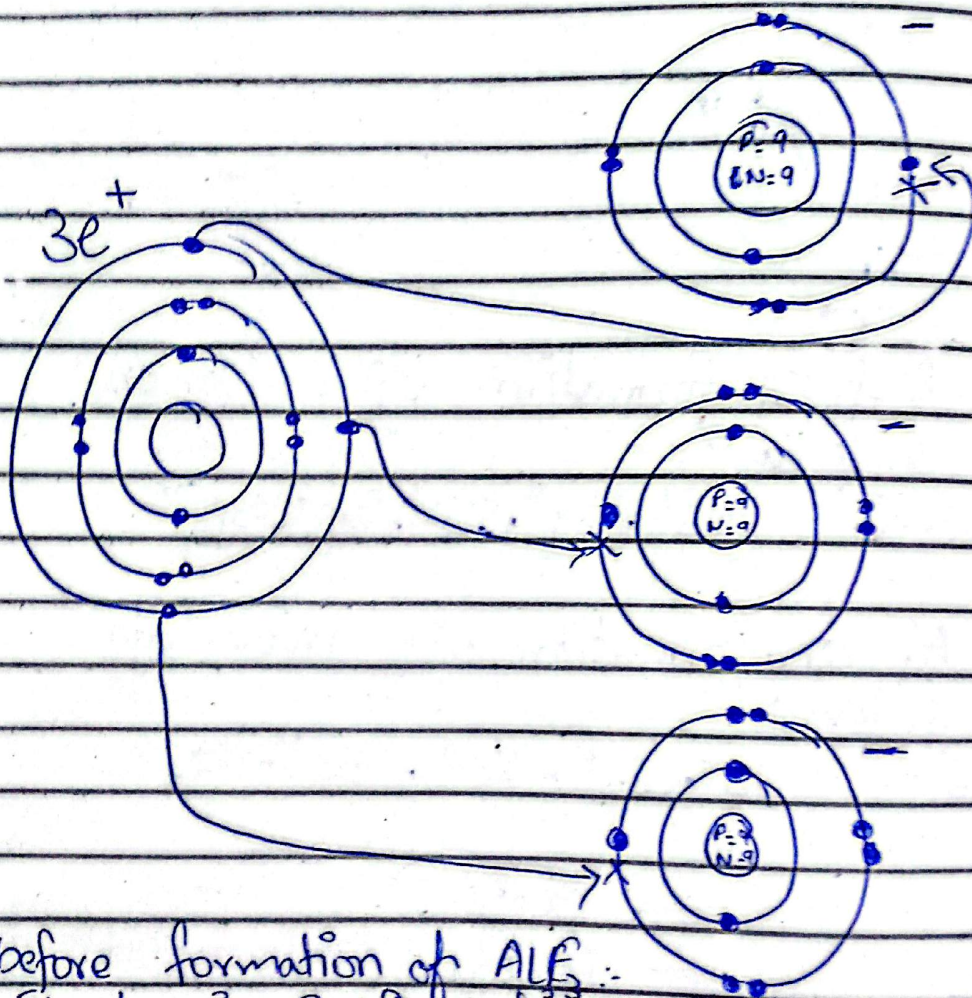
F Cg G (Li, Na) = metals

C Cg E (N, P) = non-metals

B Cg D (Ne, He) : noble gases.



$\text{Al} = 13$        $\text{F} = 9$   
 formation of  $\text{AlF}_3$  :-



before formation of  $\text{AlF}_3$  :-  
 Electronic Configuration :-

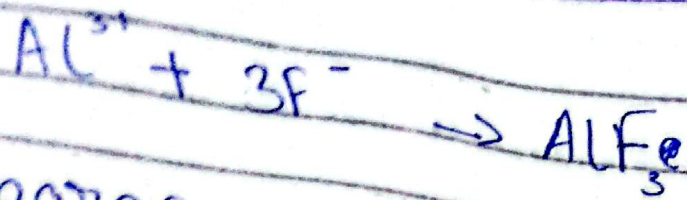
$\text{Al} : 1s^2, 2s^2, 2p^6, 3s^2, 3p^1$

$\text{F} : 1s^2, 2s^2, 2p^5$

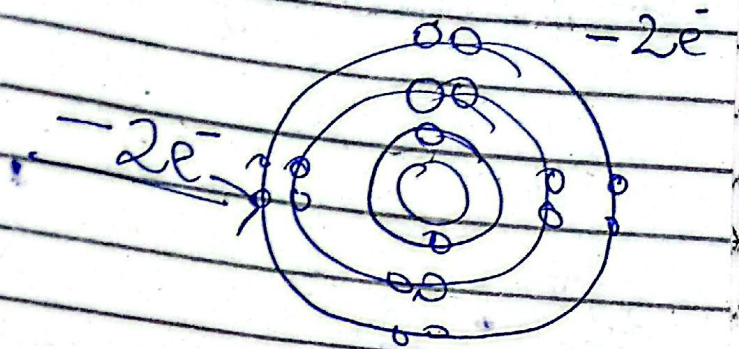
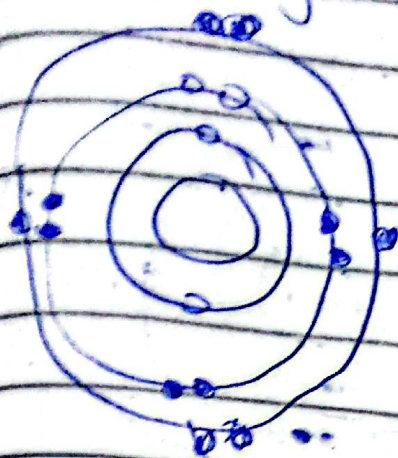
after formation of  $\text{AlF}_3$  :-  
 E.C

$\text{Al} : 1s^2, 2s^2, 2p^6$

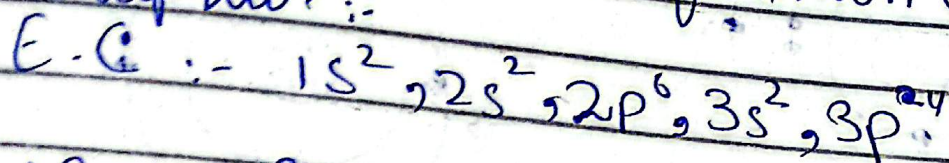
$\text{F} : 1s^2, 2s^2, 2p^6$



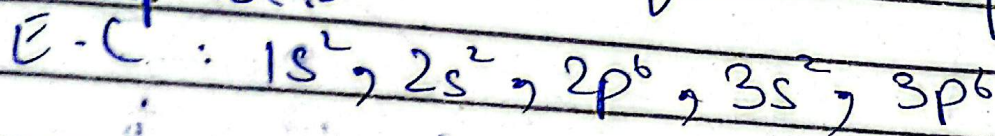
Charge on atomic n.o = 16.



Before formation of anion on sulphur:-



After formation of anion of sulphur:-



Q: 3(iv)

A: 3(iv)

a) molar mass of indigo ( $C_{16}H_{10}N_2O_2$ ):

$$= 16(12) + 10(1) + 2(14) + 2(16)$$

$$192 + 10 + 28 + 32$$

$$202 + 28 + 32$$

$$262 \text{ g/mol}$$

⑥ Molar mass of indonyl ( $C_8H_7ON$ ):

$$\begin{aligned} &= 8(12) + 7(1) + 1(16) + 1(14) \\ &\quad 92 + 7 + 16 + 14 \\ &\quad 99 + 16 + 14 \\ &\quad 129 \text{ g/mol} \end{aligned}$$

⑦ Empirical formula :-

Indigo ( $C_{16}H_{10}N_2O_2$ ) :-

$$\begin{array}{cccc} 16 & : & 10 & : & 2 & : & 2 \\ 8 & : & 5 & : & 1 & : & 1 \end{array}$$

$C_8H_5NO$  (Empirical formula)

Indonyl ( $C_8H_7ON$ )

Empirical formula :-  $C_8H_7ON$ .

Q: 2(i)

A: 2(i)

Forensic chemistry helps solve crime by analyzing physical evidence (such as blood, fibres, and trace chemicals) found at a crime scene to identify substance and link them to separate or events. It is a specific application of the broader branch analytical chemistry.

Q: 2(ii)

A: 2(ii)

Homogenous  
solutions

Heterogenous  
mixture/solution

It has uniform  
composition.

It doesn't have  
Uniform composition

It cannot be distinguished

It can be distinguished

It cannot be seen  
with naked eye

It can be seen by  
naked eye.

For example:-

For Example:-

Salt water

Fruit Salad.

Q: 2(iii) (or)

A: 2(iii) (or)

The M-shell ( $n=3$ ) contains the  
Sub shells :-

$3s, 3p, 3d$ .

In order of increasing energy :-

$3s < 3p < 3d$ .

Q: 2(iv)

A: 2(iv)

P-subshell can accommodate  
maximum of 6 electrons :-

Electronic Configuration of Sodium :-  
 $1s^2, 2s^2, 2p^6, 3s^1$ .

Q: 2(v)

A: 2(v)

Sodium has higher shielding effect than  
Lithium.

Reason:- Sodium (2, 8, 1) has more  
inner electron shells (2 complete shells)  
b/w its nucleus and valence electrons

than Lithium (2,1), which has only 1 inner shell. More inner shells means Greater shielding effects of the nuclear charge felt by the valence electron.

Q: 2(vi)

A: 2(vi)

Aluminium (Al: 13) :-

E.C :-  $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$

Group :- (since the sub-shell is p so we have to add both s and p to have the answer and sum up to 10)

$$2 + 1 = 3 + 10 = 13$$

Group 13 or Group<sup>3</sup>  
(group III A)

Potassium :- (K: 19)

E.C :-  $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$

Group :- 1 (group I A)

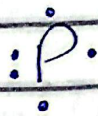
Q. 2 (vii)

A: 2 (vii)

Phosphorus (group VA) (period = 3) :-  
has 5 valence electrons and needs  
3 electrons more to complete its  
octet

$$(5 + 3 = 8)$$

Electron-dot structure :-

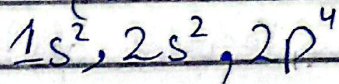


Q: 2 (viii) (or)

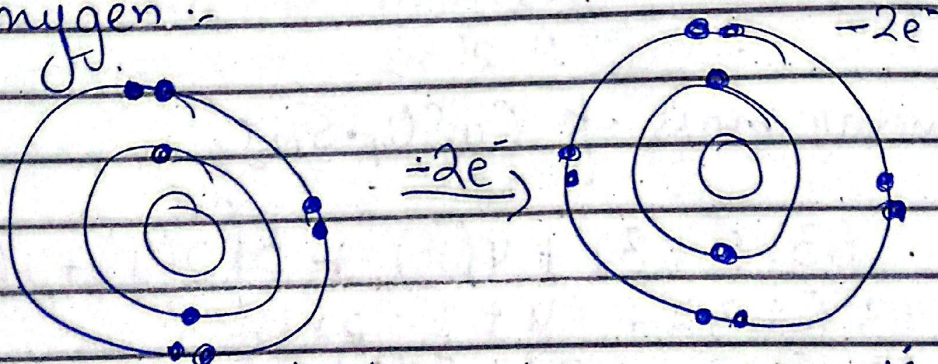
A: 2 (viii) (or)

Atomic number = 8 (oxygen) :-

E.C (Electronic Configuration) :-



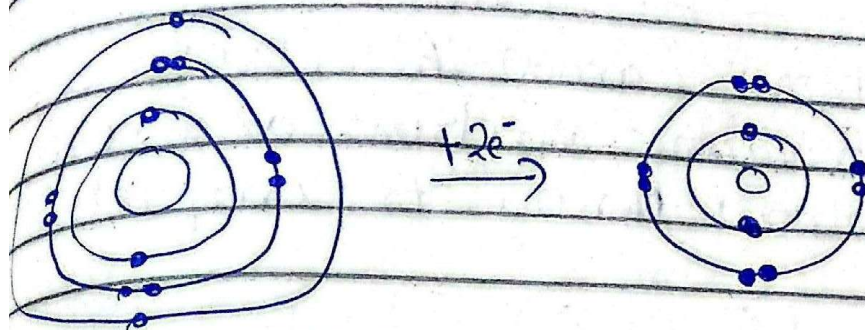
Oxygen :-



(gains 2 electrons to complete its  
octet)

E.C becomes:  $1s^2, 2s^2, 2p^6$

Atomic no. = 12 (magnesium):-  
 E.C :-  $1s^2, 2s^2, 2p^6, 3s^2$



E.C now becomes :-  
 $1s^2, 2s^2, 2p^6$

Q: 2(in)

A: 2(in)

Numbers of molecule :-

moles  $\times N_A =$

$$1.09 \times 6.022 \times 10^{23}$$

$6.564 \times 10^{23}$  molecules of benzene

Q: 2(n)

A: 2(n)

molar mass of  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$63.5 + 32 + 4(16) + 5(2(1) + 16)$$

$$63.5 + 32 + 64 + 5(2 + 16)$$

$$63.5 + 32 + 64 + 5(18)$$

$$63.5 + 32 + 64 + 90 =$$

$$249.5 \text{ g/mol}$$

$$\text{molecules} = \frac{0.5 \times 6.022 \times 10^{23}}{3.011 \times 10^{23}} \text{ molecule.}$$

Q: 2(ni)(os) —

A: 2(ni)(os) —

Q: N in  $\text{NaNO}_2$  :-

$$+1 + x + 2(-2) = 0$$

$$1 + x - 4 = 0$$

$$x - 4 + 1 = 0$$

$$x - 3 = 0$$

$$x = +3$$

MCQs :-

1. B

2. C

3. B

4. D

5. C

6. C

7. C

8. C

9. C

10. C

11. B

12. B